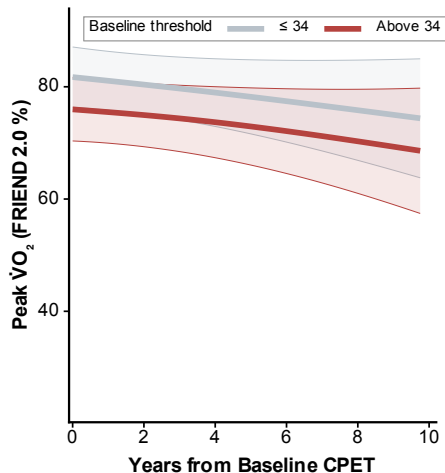
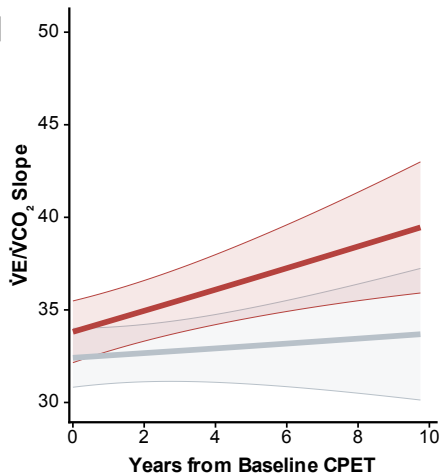


(A) Peak $\dot{V}O_2$ Baseline LAVi ≤ 34 mL/m² versus above 34 mL/m²**(B) $\dot{V}E/\dot{V}CO_2$ slope**Baseline LAVi ≤ 34 mL/m² versus above 34 mL/m²**(C) Longitudinal Model Summary**

Model component	edf	F statistic	P value
Peak $\dot{V}O_2$			
Time smooth	1.07	1.52	0.192
Threshold group	—	2.37	0.124
Time x threshold group	1.10	0.07	0.917
$\dot{V}E/\dot{V}CO_2$ slope			
Time smooth	1.04	0.43	0.512
Threshold group	—	4.54	0.034
Time x threshold group	1.04	2.84	0.089

Supplemental Figure. Parameter-specific longitudinal cardiopulmonary trajectories stratified by binary baseline LAVi (mL/m²) status. **(A)** GAMM-derived Peak $\dot{V}O_2$ trajectories. **(B)** GAMM-derived $\dot{V}E/\dot{V}CO_2$ slope trajectories. **(C)** Summary of the key status and interaction terms for both models. Baseline threshold groups were defined as LAVi ≤ 34 mL/m² versus above 34 mL/m². Longitudinal models were adjusted for age, sex, and BMI.